Toxicity Predictions (LD_{50}) & Model Interpretations for Selected Drug Combinations

This document presents a set of drug–drug combination test cases generated as part of the project "Deep Graph-based QSAR Modeling with XAI for Interpretable Prediction of Drug–Drug Interaction Toxicity (LD_{50})". For each pair, the model predicts a toxicity value and provides an explanation by highlighting important atoms and structural features in the molecules.

LD_{50} (median lethal dose) refers to the amount of a substance required to cause death in 50% of a test population, and it is commonly used as a measure of toxicity.

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Test Case 01 (Tramadol + Diazepam)

```
Resolved Drug A (name): Tramadol
SMILES A: COc1cccc(C2(O)CCCCC2CN(C)C)c1
Resolved Drug B (name): Diazepam
SMILES B: CN1C(=O)CN=C(c2ccccc2)c2cc(Cl)ccc21

--- Chemical Interpretation ---

Drug A: Tramadol
Drug B: Diazepam

Predicted pLD50: 2.2800
Predicted toxicity level: moderate toxicity

1. Model focus

Drug A important atoms:
- N sp3 degree 3 (atom 15); C sp3 degree 1 3 H (atom 0); aromatic C sp2 de
gree 3 (atom 2)

Drug B important atoms:
```

- aromatic C sp2 degree 2 1 H (atom 10); aromatic C sp2 degree 2 1 H (atom 9); aromatic C sp2 degree 2 1 H (atom 11)

2. Chemical meaning of important atoms

Drug A:
- Atom 15 is a nitrogen atom, which may indicate an amine-like or heteroat om-rich reactive region. Atom 0 is a C atom that may contribute to the loc al chemical environment. Atom 2 is an aromatic carbon, which may belong to a hydrophobic ring system.

Drug B:
- Atom 10 is an aromatic carbon, which may belong to a hydrophobic ring sy stem. Atom 9 is an aromatic carbon, which may belong to a hydrophobic ring system. Atom 11 is an aromatic carbon, which may belong to a hydrophobic r ing system.

3. Key structural fragments

Drug A:
- C -> a general structural fragment; c -> an aromatic hydrophobic ring; C NC -> an amine-like or nitrogen-containing region

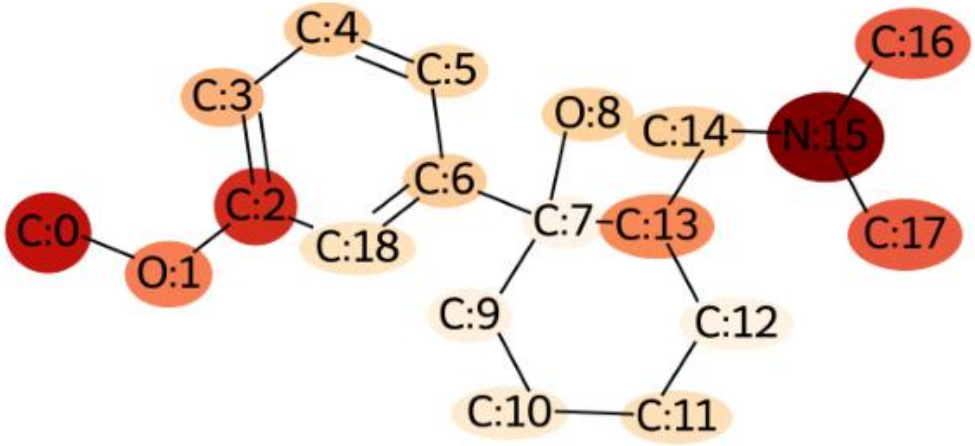
Drug B:
- N -> a nitrogen-containing heterocycle; ccc -> an aromatic hydrophobic r ing

4. Interaction-level interpretation

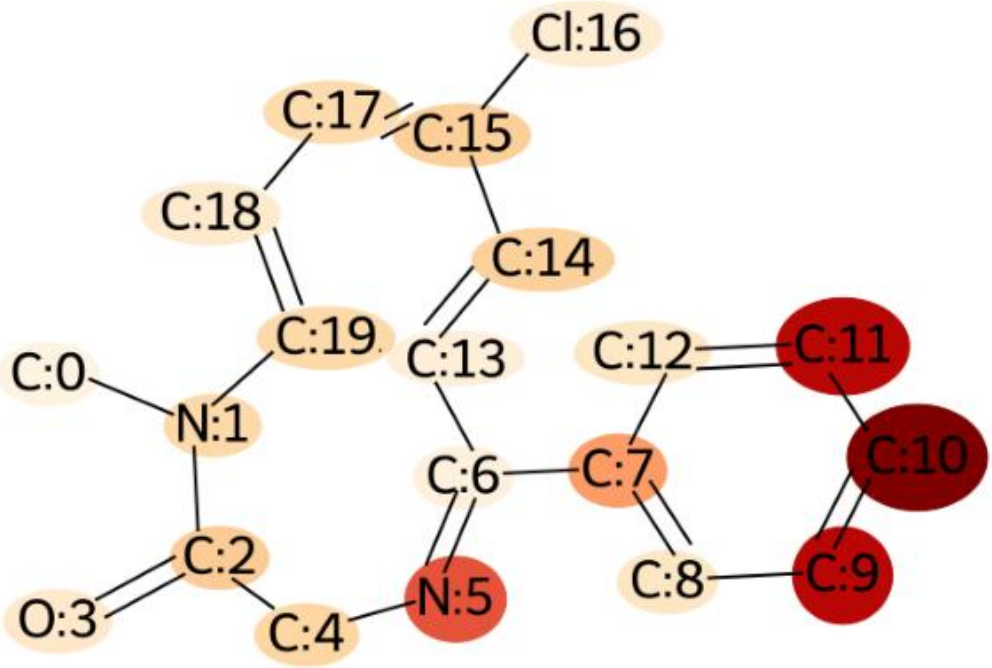
The model highlights a general structural fragment in Tramadol and a nitro gen-containing heterocycle in Diazepam. This suggests that toxicity may be influenced by the interaction between important local structural regions i n both molecules.

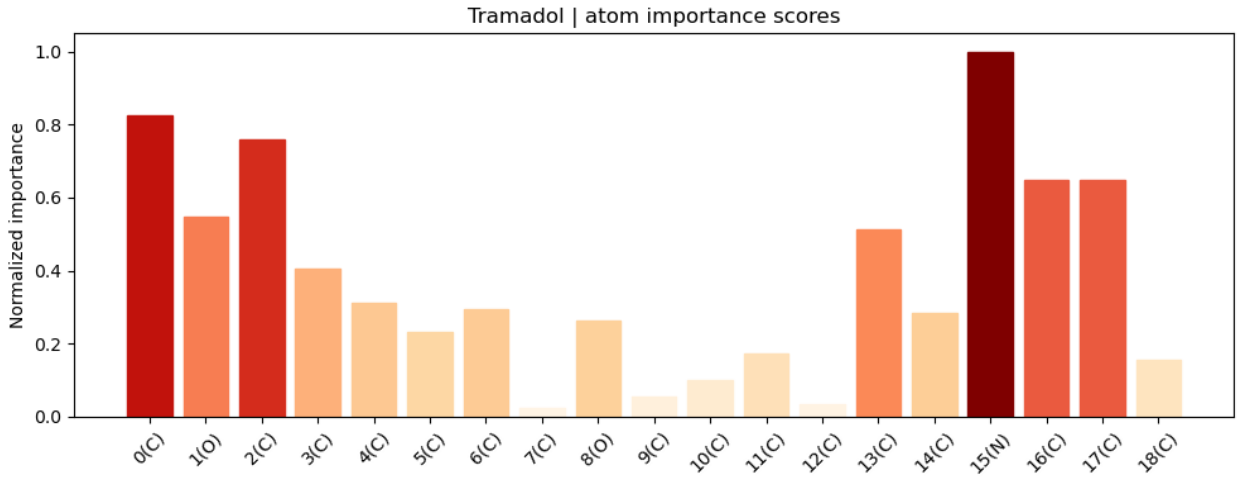
NOTE:
This explanation reflects how the model makes decisions. It does not repre sent a confirmed biological mechanism.

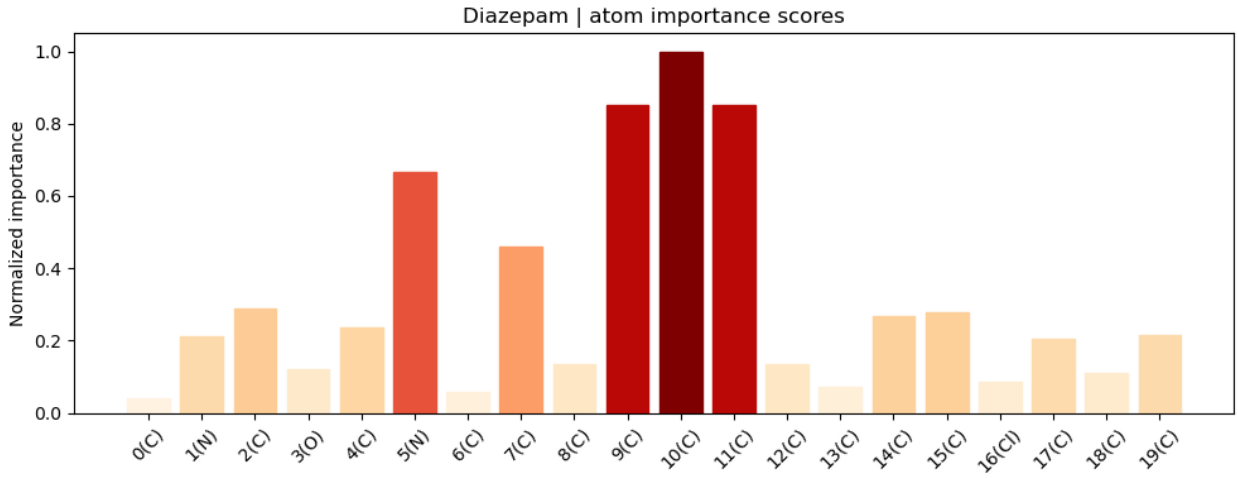
Tramadol | highlighted atom importance



Diazepam | highlighted atom importance







Test Case 02 (Paracetamol + Phenytoin)

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Resolved Drug A (smiles): N-(4-hydroxyphenyl)acetamide
SMILES A: CC(=O)Nc1ccc(O)cc1
Resolved Drug B (smiles): 5,5-diphenylimidazolidine-2,4-dione
SMILES B: O=C1NC(=O)C(c2ccccc2)(c2ccccc2)N1

--- Chemical Interpretation ---

Drug A: N-(4-hydroxyphenyl)acetamide
Drug B: 5,5-diphenylimidazolidine-2,4-dione

Predicted pLD50: 2.6200
Predicted toxicity level: high toxicity

1. Model focus
```

Drug A important atoms:
- aromatic C sp2 degree 3 (atom 7); O sp2 degree 1 1 H (atom 8); N sp2 degree 2 1 H (atom 3)

Drug B important atoms:
- ring N sp2 degree 2 1 H (atom 18); ring C sp3 degree 4 (atom 5); ring N sp2 degree 2 1 H (atom 2)

2. Chemical meaning of important atoms

Drug A:
- Atom 7 is an aromatic carbon, which may belong to a hydrophobic ring system. Atom 8 is an oxygen atom, which may indicate a polar functional group. Atom 3 is a nitrogen atom, which may indicate an amine-like or heteroatom-rich reactive region.

Drug B:
- Atom 18 is a ring nitrogen, which may influence polarity, basicity, or binding behavior. Atom 5 is a ring carbon, which may contribute to structural rigidity. Atom 2 is a ring nitrogen, which may influence polarity, basicity, or binding behavior.

3. Key structural fragments

Drug A:
- cO -> an aromatic hydrophobic ring

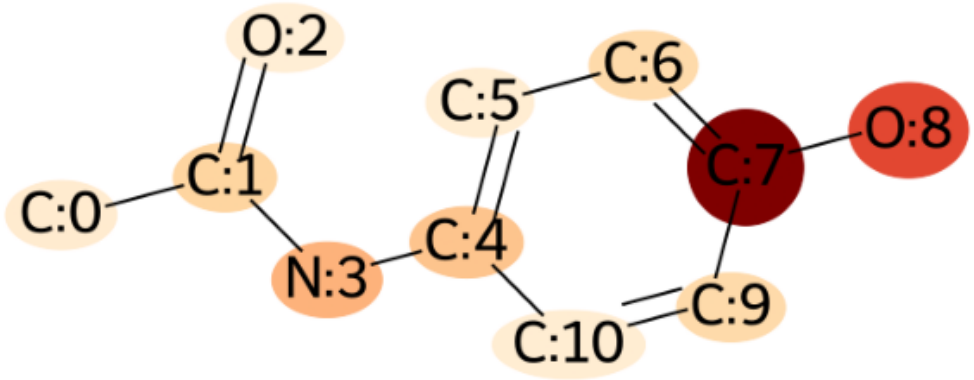
Drug B:
- CN -> a nitrogen-containing heterocycle

4. Interaction-level interpretation

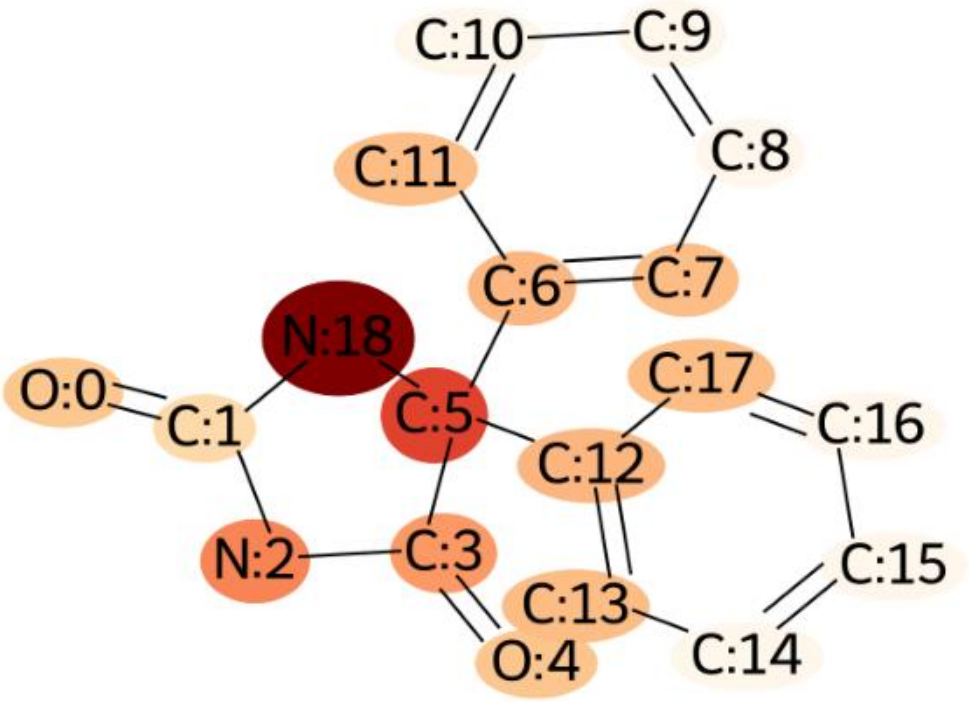
The model highlights an aromatic hydrophobic ring in N-(4-hydroxyphenyl)acetamide and a nitrogen-containing heterocycle in 5,5-diphenylimidazolidine-2,4-dione. This suggests that toxicity may be influenced by the interaction between important local structural regions in both molecules.

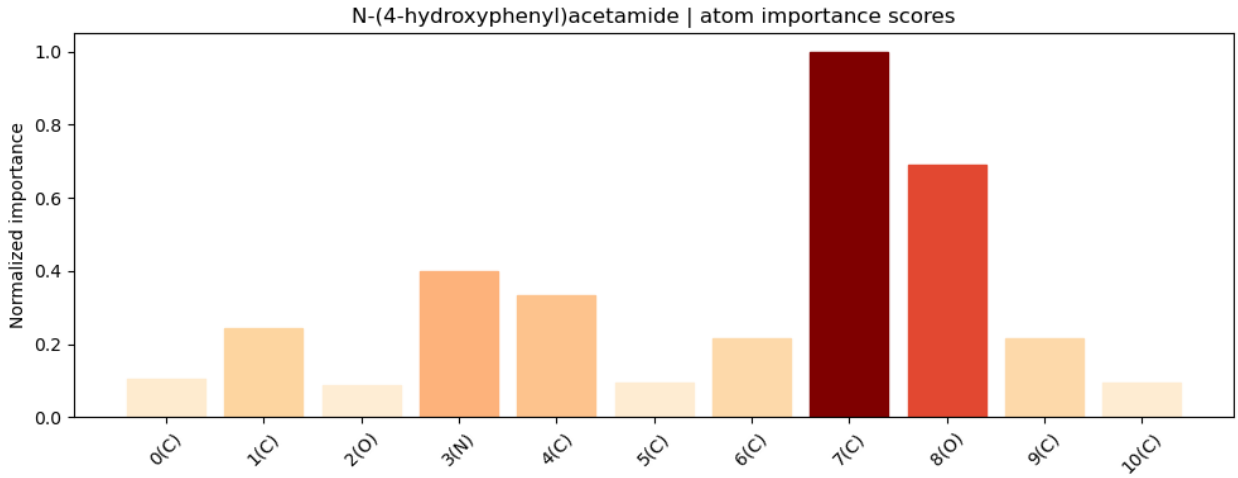
NOTE:
This explanation reflects how the model makes decisions. It does not represent a confirmed biological mechanism.

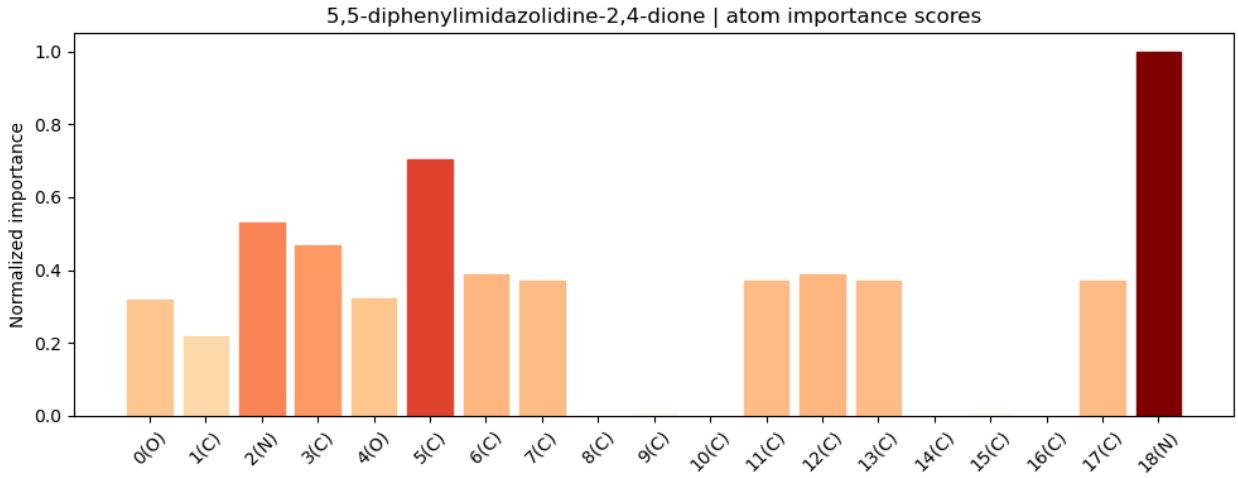
N-(4-hydroxyphenyl)acetamide | highlighted atom importance



5,5-diphenylimidazolidine-2,4-dione | highlighted atom importance







Test Case 03 (Warfarin + Ibuprofen)

```
Enter Drug A common name or SMILES: Warfarin
Enter Drug B common name or SMILES: Ibuprofen
Resolved Drug A (name): Warfarin
SMILES A: CC(=O)CC(c1ccccc1)c1c(O)c2ccccc2oc1=O
Resolved Drug B (name): Ibuprofen
SMILES B: CC(C)Cc1ccc(C(C)C(=O)O)cc1

--- Chemical Interpretation ---

Drug A: Warfarin
Drug B: Ibuprofen

Predicted pLD50: 2.5086
Predicted toxicity level: high toxicity

1. Model focus
```

Drug A important atoms:
- C sp3 degree 3 1 H (atom 4); O sp2 degree 1 (atom 22); aromatic C sp2 degree 2 1 H (atom 8)

Drug B important atoms:
- aromatic C sp2 degree 3 (atom 7); C sp3 degree 2 2 H (atom 3); C sp3 degree 3 1 H (atom 1)

2. Chemical meaning of important atoms

Drug A:
- Atom 4 is a highly connected C atom, which may lie in a structurally important region. Atom 22 is an oxygen atom, which may indicate a polar functional group. Atom 8 is an aromatic carbon, which may belong to a hydrophobic ring system.

Drug B:
- Atom 7 is an aromatic carbon, which may belong to a hydrophobic ring system. Atom 3 is a C atom that may contribute to the local chemical environment. Atom 1 is a highly connected C atom, which may lie in a structurally important region.

3. Key structural fragments

Drug A:
- C=O -> an oxygen-rich polar chain or functional group; C -> a general structural fragment; ccc -> an aromatic hydrophobic ring; O -> an oxygen-rich polar chain or functional group

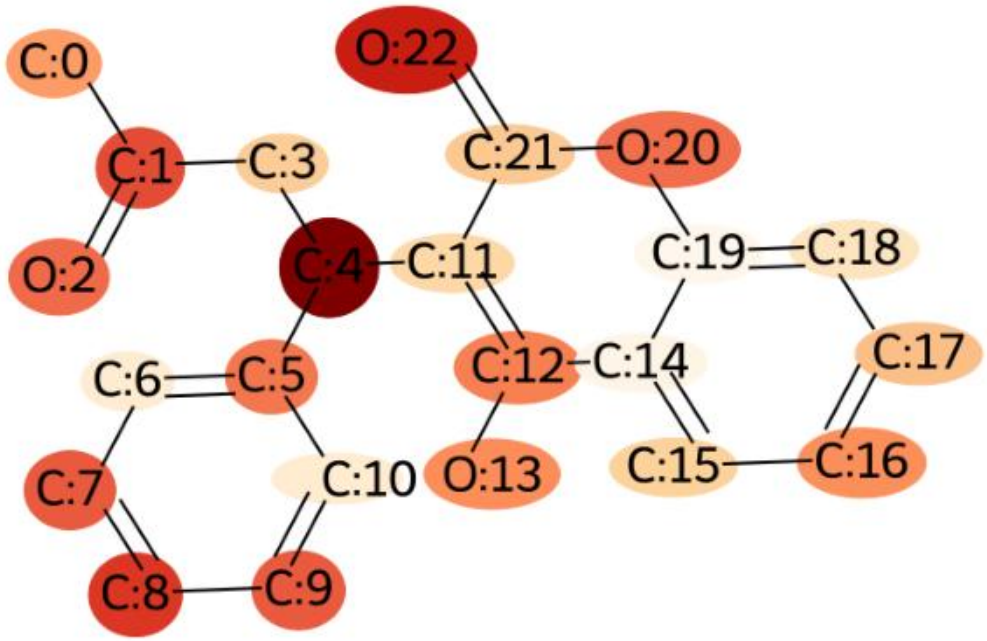
Drug B:
- CC -> a general structural fragment; c -> an aromatic hydrophobic ring

4. Interaction-level interpretation

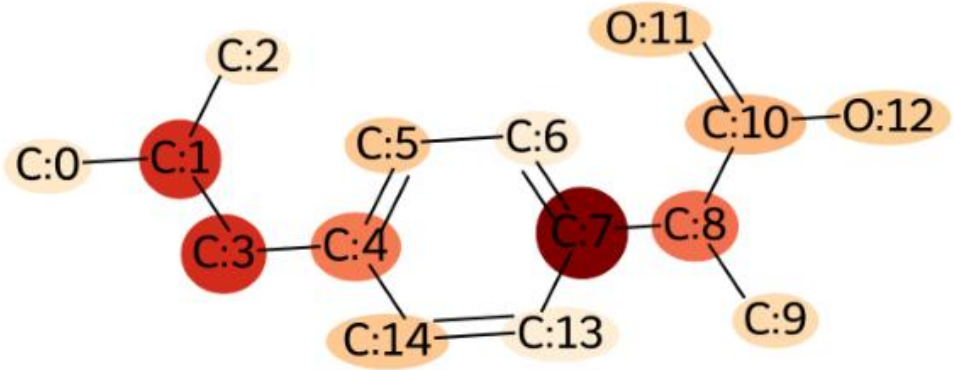
The model highlights an oxygen-rich polar chain or functional group in Warfarin and a general structural fragment in Ibuprofen. This suggests that toxicity may be influenced by the interaction between important local structural regions in both molecules.

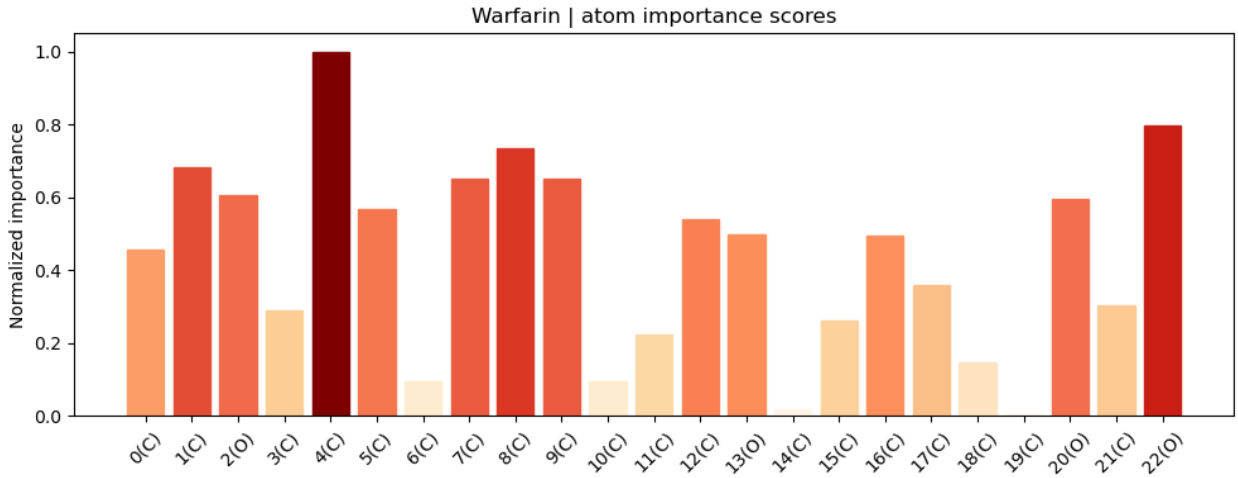
NOTE:
This explanation reflects how the model makes decisions. It does not represent a confirmed biological mechanism.

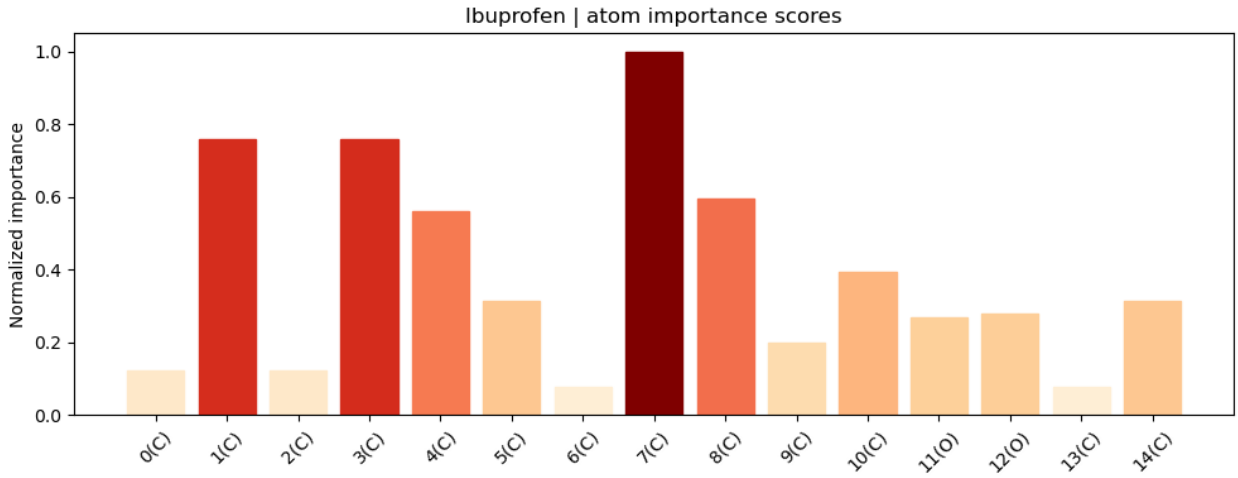
Warfarin | highlighted atom importance



Ibuprofen | highlighted atom importance







Test Case 04 (Vitamin C + Paracetamol)

```
Enter Drug A common name or SMILES: Vitamin C
Enter Drug B common name or SMILES: Paracetamol
Resolved Drug A (name): Vitamin C
SMILES A: O=C1OC(C(O)CO)C(O)=C1O
Resolved Drug B (name): Paracetamol
SMILES B: CC(=O)Nc1ccc(O)cc1

--- Chemical Interpretation ---

Drug A: Vitamin C
Drug B: Paracetamol

Predicted pLD50: 1.7445
Predicted toxicity level: low toxicity

1. Model focus
```

Drug A important atoms:
- C sp3 degree 2 2 H (atom 6); ring C sp2 degree 3 (atom 1); ring C sp3 degree 3 1 H (atom 3)

Drug B important atoms:
- O sp2 degree 1 1 H (atom 8); C sp3 degree 1 3 H (atom 0); C sp2 degree 3 (atom 1)

2. Chemical meaning of important atoms

Drug A:
- Atom 6 is a C atom that may contribute to the local chemical environment. Atom 1 is a ring carbon, which may contribute to structural rigidity. Atom 3 is a ring carbon, which may contribute to structural rigidity.

Drug B:
- Atom 8 is an oxygen atom, which may indicate a polar functional group. Atom 0 is a C atom that may contribute to the local chemical environment. Atom 1 is a highly connected C atom, which may lie in a structurally important region.

3. Key structural fragments

Drug A:
- C -> a non-aromatic ring structure; C -> a general structural fragment

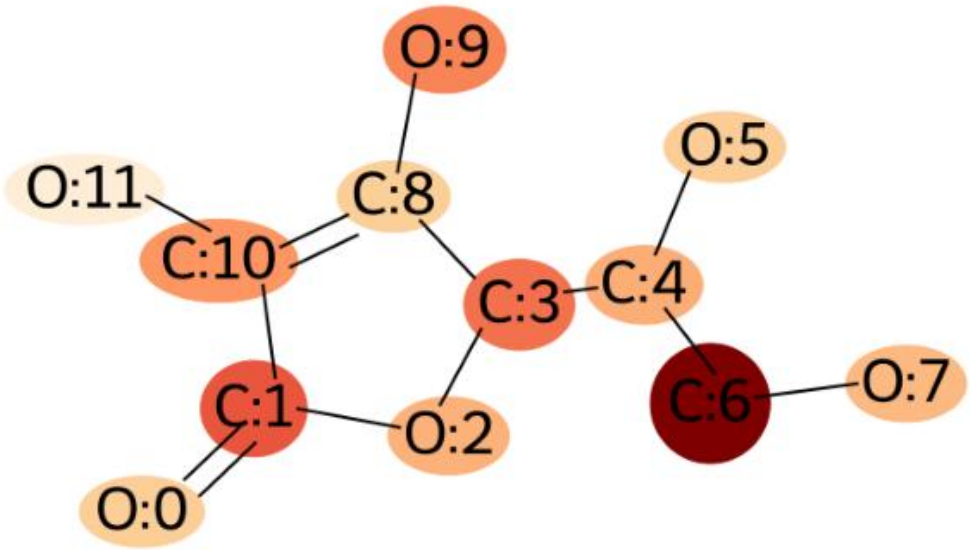
Drug B:
- O -> an oxygen-rich polar chain or functional group

4. Interaction-level interpretation

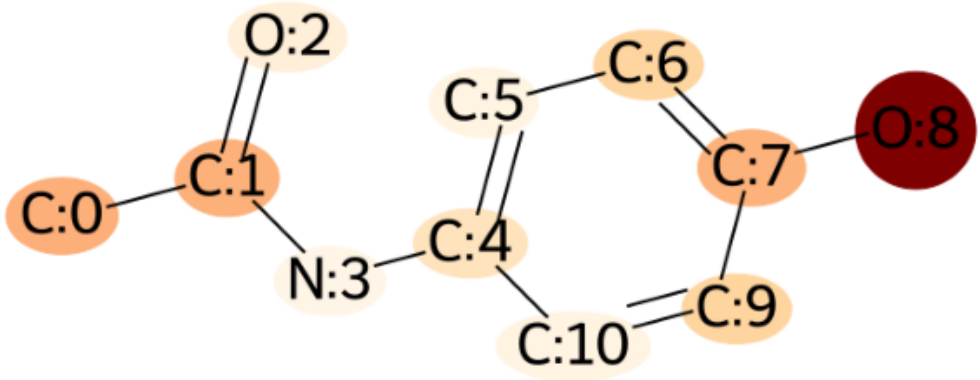
The model highlights a non-aromatic ring structure in Vitamin C and an oxygen-rich polar chain or functional group in Paracetamol. This suggests that toxicity may be influenced by the interaction between important local structural regions in both molecules.

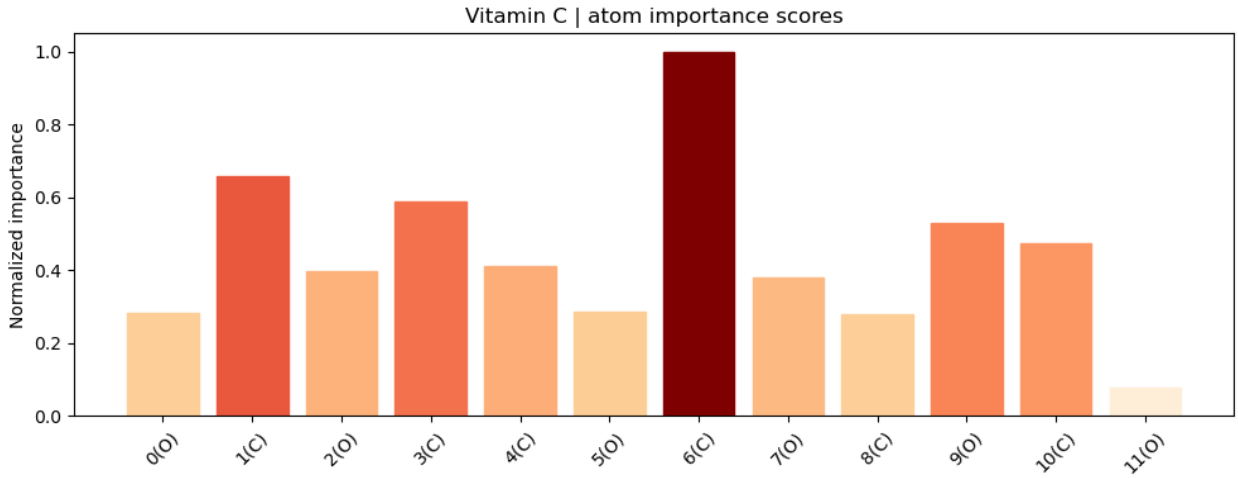
NOTE:
This explanation reflects how the model makes decisions. It does not represent a confirmed biological mechanism.

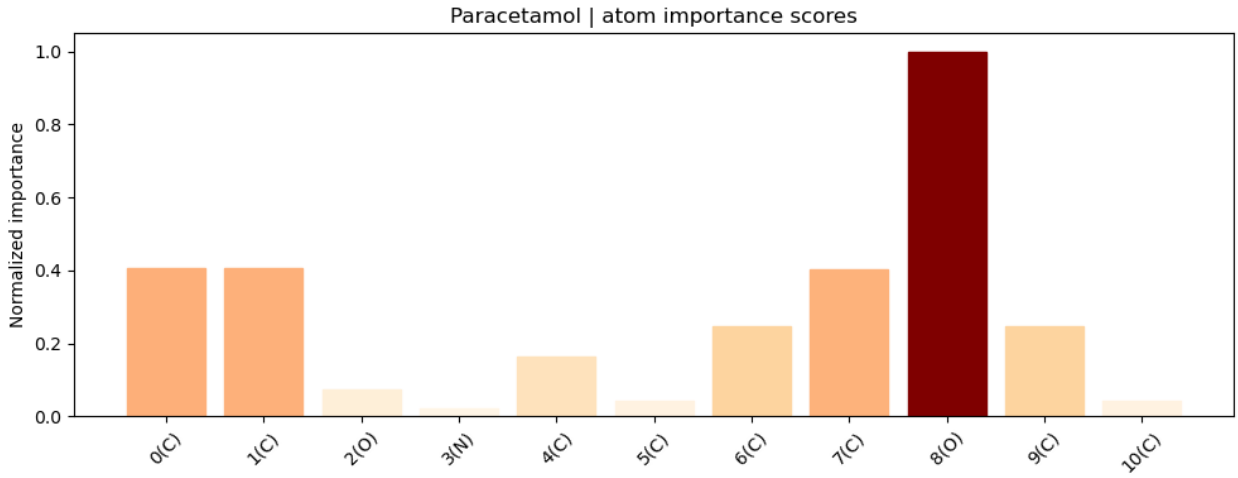
Vitamin C | highlighted atom importance



Paracetamol | highlighted atom importance







Test Case 05 (Tramadol + Sertraline)

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Enter Drug A common name or SMILES: Tramadol
Enter Drug B common name or SMILES: Sertraline
Resolved Drug A (name): Tramadol
SMILES A: COc1cccc(C2(O)CCCCC2CN(C)C)c1
Resolved Drug B (name): Sertraline
SMILES B: CNC1CCC(c2ccc(Cl)c(Cl)c2)c2ccccc21

--- Chemical Interpretation ---

Drug A: Tramadol
Drug B: Sertraline

Predicted pLD50: 2.6709
Predicted toxicity level: high toxicity

1. Model focus
```

Drug A important atoms:
- N sp3 degree 3 (atom 15); O sp2 degree 2 (atom 1); ring C sp3 degree 2 2 H (atom 12)

Drug B important atoms:
- ring C sp3 degree 3 1 H (atom 2); ring C sp3 degree 2 2 H (atom 3); N sp 3 degree 2 1 H (atom 1)

2. Chemical meaning of important atoms

Drug A:
- Atom 15 is a nitrogen atom, which may indicate an amine-like or heteroat om-rich reactive region. Atom 1 is an oxygen atom, which may indicate a po lar functional group. Atom 12 is a ring carbon, which may contribute to st ructural rigidity.

Drug B:
- Atom 2 is a ring carbon, which may contribute to structural rigidity. At om 3 is a ring carbon, which may contribute to structural rigidity. Atom 1 is a nitrogen atom, which may indicate an amine-like or heteroatom-rich re active region.

3. Key structural fragments

Drug A:
- O -> an oxygen-rich polar chain or functional group; N -> an amine-like or nitrogen-containing region

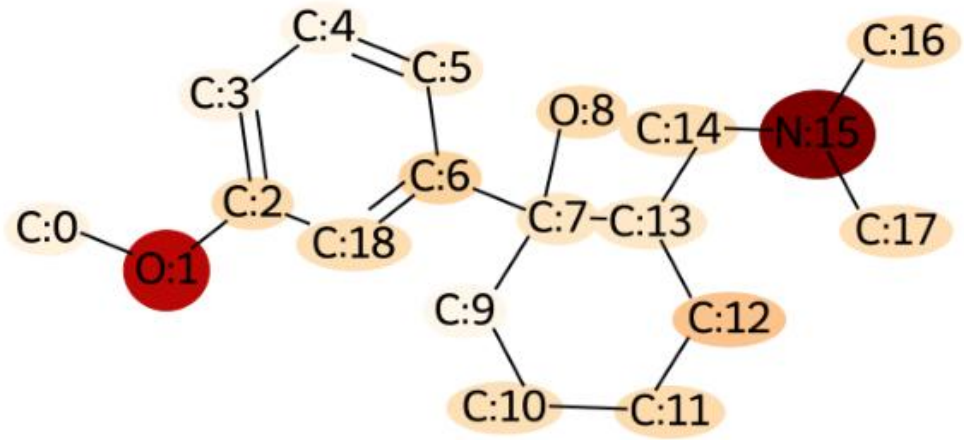
Drug B:
- C -> a non-aromatic ring structure

4. Interaction-level interpretation

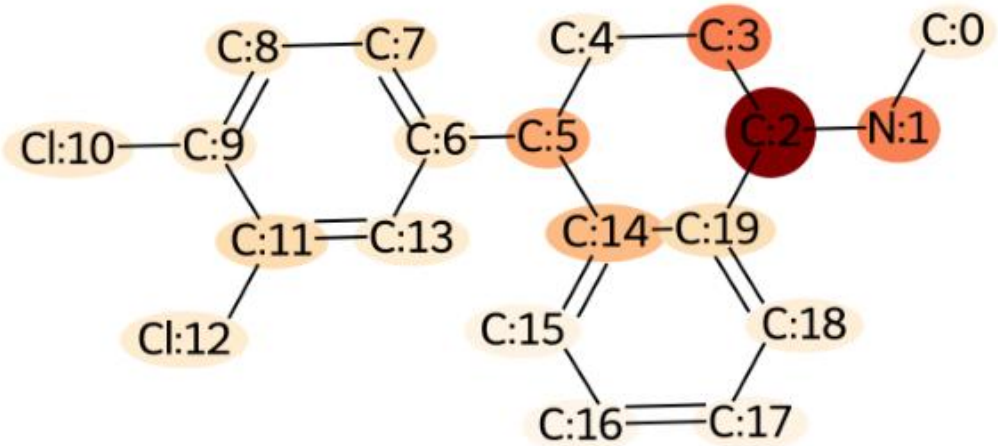
The model highlights an oxygen-rich polar chain or functional group in Tra madol and a non-aromatic ring structure in Sertraline. This suggests that toxicity may be influenced by the interaction between important local stru ctural regions in both molecules.

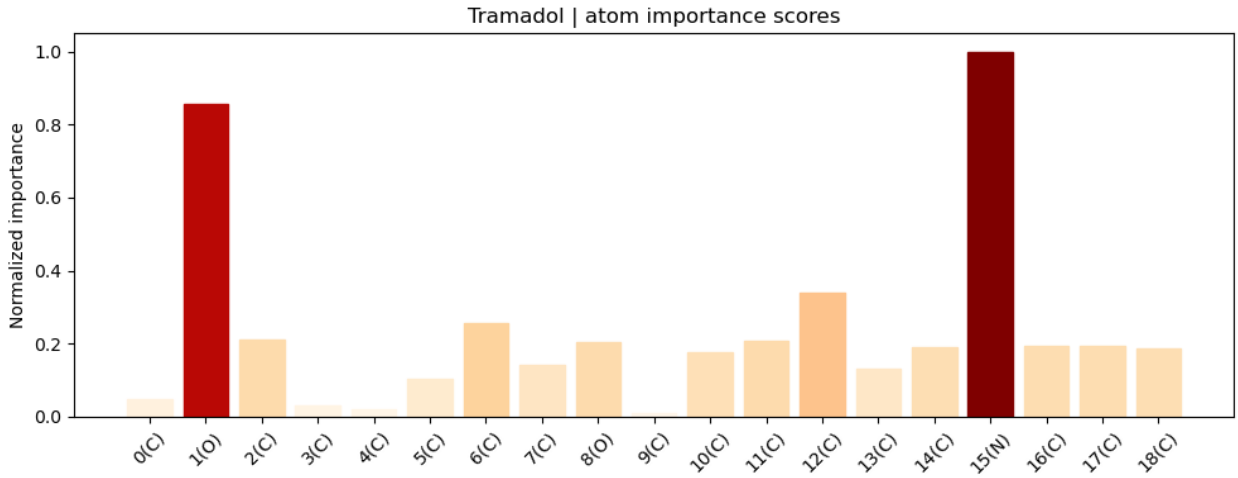
NOTE:
This explanation reflects how the model makes decisions. It does not repre sent a confirmed biological mechanism.

Tramadol | highlighted atom importance



Sertraline | highlighted atom importance




Tramadol | atom importance scores
Normalized importance
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0.8
0.6
0.4
0.2
0.0
0(C)
1(O)
2(C)
3(C)
4(C)
5(C)
6(C)
7(C)
8(O)
9(C)
10(C)
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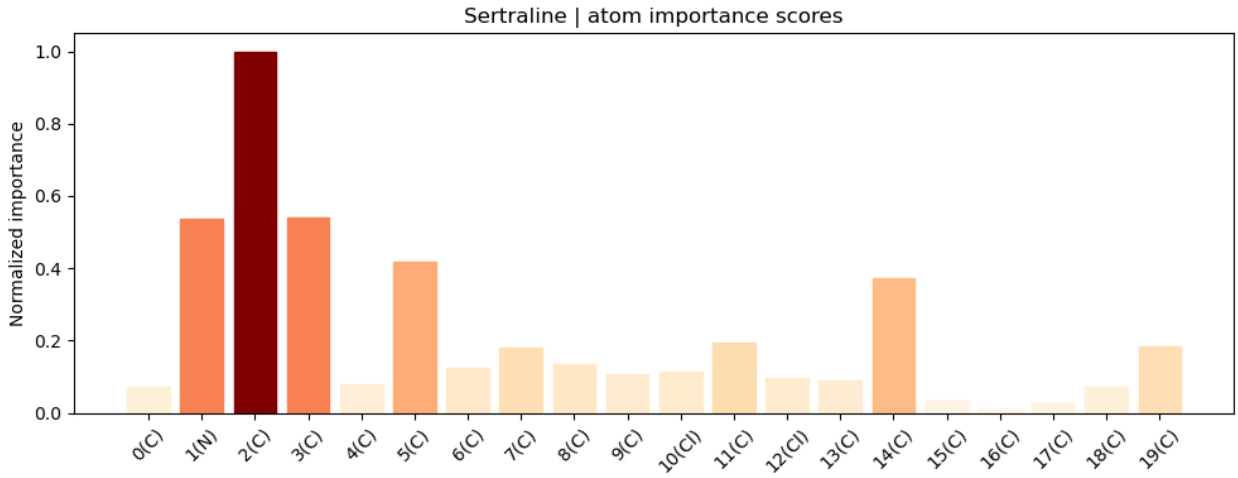
14(C)
15(N)
16(C)
17(C)
18(C)


Sertraline | atom importance scores
Normalized importance
1.0
0.8
0.6
0.4
0.2
0.0
0(C)
1(N)
2(C)
3(C)
4(C)
5(C)
6(C)
7(C)
8(C)
9(C)
10(Cl)
11(C)
12(Cl)
13(C)
14(C)
15(C)
16(C)
17(C)
18(C)
19(C)